



Unleashing the Power of Event-Driven Integrations with Advanced Event Mesh

SAP TECHNICAL ACADEMY

THE BEST RUN 

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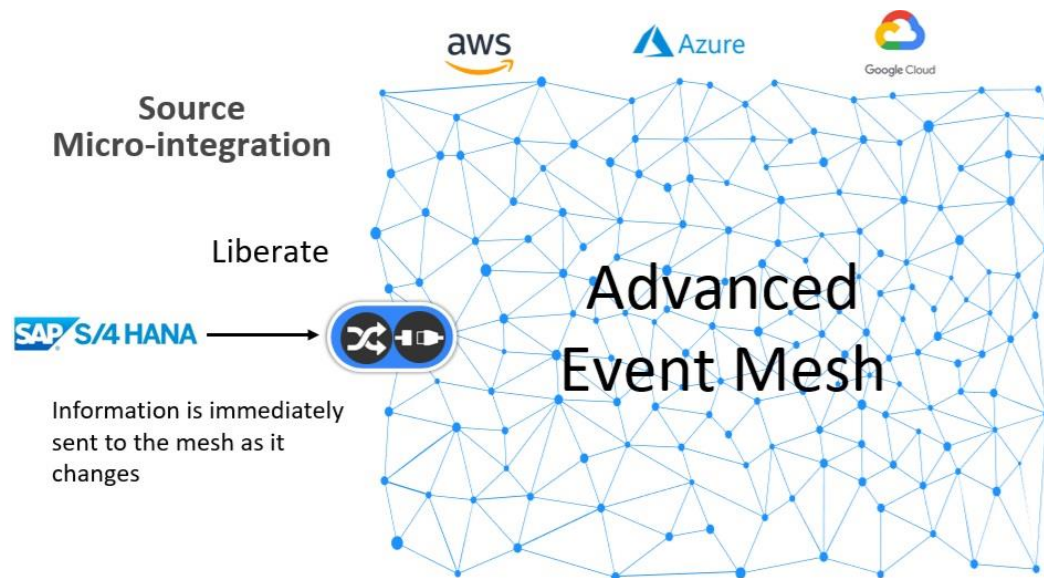
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1 Introduction to Smart Topics

1.1 Setting up our Environment

We will be simulating a typical customer setup where Events are emitted from SAP S/4 or ECC system and published to the Advanced Event Mesh. This allows customers to react in real time to these events with services like SAP Business Process Automation or Cloud Integration.

The pattern that we suggest to customers is the **Micro Integration Pattern** where interfaces are established without any knowledge of the applications that will consume them. In the diagram below, the destination for our events is the Advanced Event Mesh(aka AEM). Events like Business Partner Create/Change will be published to AEM without any knowledge of which or how many applications will be using those events. This simplifies integration as you do not require knowledge of the consuming applications.



In our setup for the workshop, we will be using an Event Generator that will already be connected to the broker that you are using today. We have activated Business Partner Change and Business Partner Create and these will be the 2 events simulated from an ERP perspective.

For example, if you had a website or vendor portal that allows your customers to change their address, you might want to publish that Address Change as an event so you could send that customer an immediate email to confirm the change was made and the address reflects accurately what the customer wanted.

We will use this setup to help you understand how to subscribe dynamically to this information coming out of SAP.

1.2 TOPICS 101

As we walk into the next section, it's useful to understand the basic structure of a topic and how it should be constructed. The best practice we recommend is to use a "NOUN", followed by "VERB" (e.g. SALESORDER/CREATED), then a set of properties that you choose depending on how you would like to access the data later.

In the diagram below, all events published related to business partner would have the first section of their Topic as "sap/erp/businesspartner/create/v1" (the static portion) and what follows would be the dynamic properties.

Properties for a business partner could include city/state/zip code, lat, long, etc...this would allow you to later subscribe and filter on customers in a certain city or geographic location etc.

Additionally, the Advanced Event Mesh service allows the use of wildcards – as seen below.

Everything that occurs is an event

What if you could "tune in" to specific kinds of events occurring across your enterprise and across clouds?

Hashtags Structure

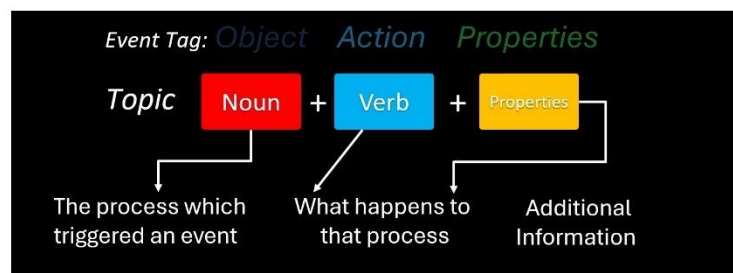
#icecream

#selfie

#crypto



Events and Topics Structure



Publish (Dynamic Topics)

- `sap/erp/businesspartner/create/v1/>`
- `sap/erp/businesspartner/change/v1/>`

Subscribe

- `>`
- `sap/erp/businesspartner/change/>`
- `sap/erp/businesspartner/change/Executive/*`

1.3 Hands On with AEM and Topics

1.3.1 Introduction to Wild Cards

The events that are being generated from the Simulator have the following topic structure:

sap/erp/businesspartner/change/V1/{businessPartnerType}/{partnerId}

sap/erp/businesspartner/create/V1/{businessPartnerType}/{partnerId}

There are 2 wildcards that can be used in filtering expressions.

1.3.1.1

‘*’ is used to match pieces of data at a certain level. For example, if I wanted all events that matched any Version number I could create a subscription that looks like this:

sap/erp/businesspartner/change/*/Executive/50

In this example, the asterisk is being used to attract events that have any version number and also just Executives with the ID of 50. Important to note that the asterisk only matches that level.

1.3.1.2

‘>’ is used to match any and all levels that follow it.

sap/erp/>

A subscription with a filtering expression like this would attract all events that have a topic starting with sap/erp, and it would match all business partners, sales orders, purchase orders, etc...

The interesting thing is that you can use wildcards in combination with each other so you can have something like:

sap/erp/*/change/>

What exactly would this Subscription attract?

2 Accessing the environment

Please use incognito / private mode to connect ... otherwise, there could be confusion with existing SAP users and you may end up in somebody else's environment!

Advanced Event Mesh

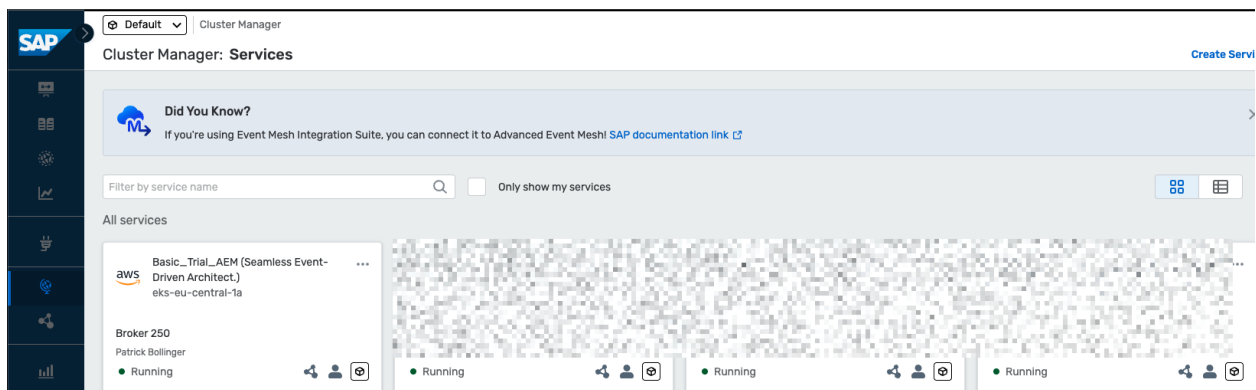
<https://eu10.console.pubsub.em.services.cloud.sap/login?tenant-id=b93dd091-7f5b-4331-9c31-3d18b38e9cce>

When you click on the link above, (if you're using a private/incognito window) you will be prompted for a username and password. If you're not prompted for a username and password, it means you're logged on to your corporate BTP account!

Username: AC373526U**nn**(Replace "**nn**" with the 2 digit ID assigned to you)

Password: <<<This will be provided to you during the hands-on exercise>>>

Once logged in, you will see the AEM console as shown in the screenshot below. You will see several AEM instances. The **AEM instance assigned to this workshop** is **Basic_Trial_AEM**



The URL for Integration Suite is shown below, and the username and password details are the same as for Advanced Event Mesh. **Remember to use an incognito window.**

Integration Suite

<https://integration-suite-us10.integrationsuite.cfapps.us10-002.hana.ondemand.com/>

2.1.1 Subscription Challenge #1

Your first challenge is to write a **subscription filtering expression** which will **attract every event** being published to the AEM service.

We will use the “TRY ME” utility for these subscription exercises. Instructions for setting up the TryMe utility are contained in the Appendix. For ease, the hyperlink is here -> [APPENDIX – CONNECTING TO THE TRY ME UTILITY](#)

Once you have successfully connected to the TRY ME facility as per the instructions located in the Appendix, enter a “>” into the “Topic Subscriber” input box and press enter.

The screenshot shows the TryMe utility interface with two main panels: Publisher and Subscriber. The Publisher panel on the left includes a 'Publish' section with a 'Topic' radio button selected, a 'Topic Structure' section with a visual builder showing 'noun / verb / property / property / property', and a 'Create a topic using the topic builder' section with input fields for 'flight', 'boarding', 'R1234', 'yow', and 'ewr'. The Subscriber panel on the right includes a 'Subscribe' section with a 'Topic Subscriber' input box containing a greater-than sign (>). Below the input box is a 'Subscribe' button. The interface also shows a 'Send and Receive' status bar at the top and a 'montrealbroker-10-1-...' user indicator.

Once you have subscribed as shown in the next screenshot, you should immediately see events appearing in the window towards the bottom of the screen. If so, you have dynamically subscribed to the mesh and had your first experience with using a Topic Subscription. It’s REALLY important to recognize that the TRY ME tab is not doing the filtering here, is the AEM service. AEM is only sending events to your “Try Me” tab that match the subscription you have entered.

This screenshot shows the TryMe utility interface after a subscription. The Subscriber panel on the right now includes a 'Subscribed Topics' section with a red box around it, showing a greater-than sign (>) and a close button (X). Below this is a 'Bind to an endpoint to receive guaranteed messages' section. The 'Messages (Most Recent 20)' section at the bottom shows a list of messages, with the first message highlighted by a red box. The message details are as follows:

Message ID	Delivery Mode	DHO Eligible	Time To Live (sec)	Message Content
2025-07-23 11:32:39.701	Direct	Yes	0	[{"businessPartner":{"firstName":"Janelle","lastName":"Smitham","addressLink":{"address":{"country":"AU","nation":"AK","city":"South Litzy","street":"Yost Squares","postalCode":"17569","houseNumber":"2438"},"addressNumber":"449","dateFrom":"Tue Jul 22 2

2.1.2 Subscription Challenge #2

Your second challenge is to write a subscription filtering expression which will attract any **Business Partner** that has been **created** or **changed** in the SAP System. As a reminder, the structure of events being published is:

sap/erp/businesspartner/change/V1/{businessPartnerType}/{partnerId}

sap/erp/businesspartner/create/V1/{businessPartnerType}/{partnerId}

Resist the temptation to look below this line and see the answer 😊

The screenshot displays the SAP Event Mesh interface, divided into two main sections: Publisher and Subscriber.

Publisher Section:

- Establish Connection:** Shows a status of "Connected" with a "Disconnect" button.
- Publish:** Includes a "Topic" radio button and a "Queue" radio button.
- Learn about topic taxonomy:** A link to learn more about best practices.
- Think Events!** A section for building smart topics.
- Topic Structure:** A visual representation of a topic structure: `noun / verb / property / property / property`.
- Create a topic using the topic builder:** A form with fields for Noun, Verb, Property, Property, and Property. The values entered are: `flight`, `boarding`, `fl1234`, `yow`, and `ewr`.
- Publish to topic:** A text field containing the topic: `flight/boarding/fl1234/yow/ewr`.
- Delivery Mode:** A section for configuring delivery mode.

Subscriber Section:

- Suggestions:** A list of suggested topics: `flight / >`, `flight / boarding / * / yow / >`, and `flight / * / * / yow / >`.
- Topic Subscriber:** A text field for entering a topic subscriber.
- Subscribe:** A button to subscribe to the topic.
- Subscribed Topics:** A list of subscribed topics: `sap/erp/businesspartner/>`.
- Bind to an endpoint to receive guaranteed messages:** A link to bind to an endpoint.
- Messages (Most Recent 20):** A list of messages. The first message is: `2025-07-23 11:37:30.690 [Topic sap/erp/businesspartner/create/V1/Architect/68]`.

So, if you remove the previous subscription ">" and enter a new subscription with this format :

sap/erp/businesspartner/>

You should see all created and changed business partners arrive in the window.

2.1.3 Subscription Challenge #3

Create a subscription that only attracts **Business Partners** that have **changed** with a business partner type of “**Developer**”.

The screenshot displays the SAP Send and Receive configuration interface, divided into two main panels: Publisher and Subscriber.

Publisher Panel:

- Establish Connection:** Shows a status of "Connected" with a "Disconnect" button.
- Publish:** Includes a "Show Advanced" link and a "Learn about topic taxonomy" link.
- Think Event!** A section with a "Topic Structure" showing a hierarchy: noun / verb / property / property / property. Below this, a "Create a topic using the topic builder" section shows a table with columns: Noun, Verb, Property, Property, Property. The table contains the following data:

Noun	Verb	Property	Property	Property
flight	/	boarding	/	11234
			/	view
				event
- Publish to topic:** A text field containing "flight/boarding/11234/view/event".
- Delivery Mode:** Radio buttons for "Direct" (selected) and "Persistent".
- Message Content:** A text area containing a JSON message:

```
{  "airline": "ExampleAirline",  "region": "Central",  "requestor": "44556",  "flight": {    "flightNumber": "boeing737"  }}
```
- Buttons:** "Publish" and "Generate Topic".

Subscriber Panel:

- Subscribe:** A section with a "Subscribe" button.
- Time into Event!** A section with a "Starts with" field containing "noun" and a "Suggestions" field containing "flight / boarding / 11234 / view / event".
- Topic Subscriber:** A text field containing "sap/erp/businesspartner/change/V1/Developer/>".
- Send to an endpoint to receive guaranteed messages:** A section with a "Messages (Most Recent 20)" list. The list contains one message:

```
2025-07-25 12:03:21:007 [Topic: sap/erp/businesspartner/change/V1/Developer/995]
```

Remove the previous subscription (sap/erp/businesspartner/>) and replace it with the following subscription:

sap/erp/businesspartner/change/V1/Developer/>

You should only see events coming into the Try Me window that are changed developer business partner records.

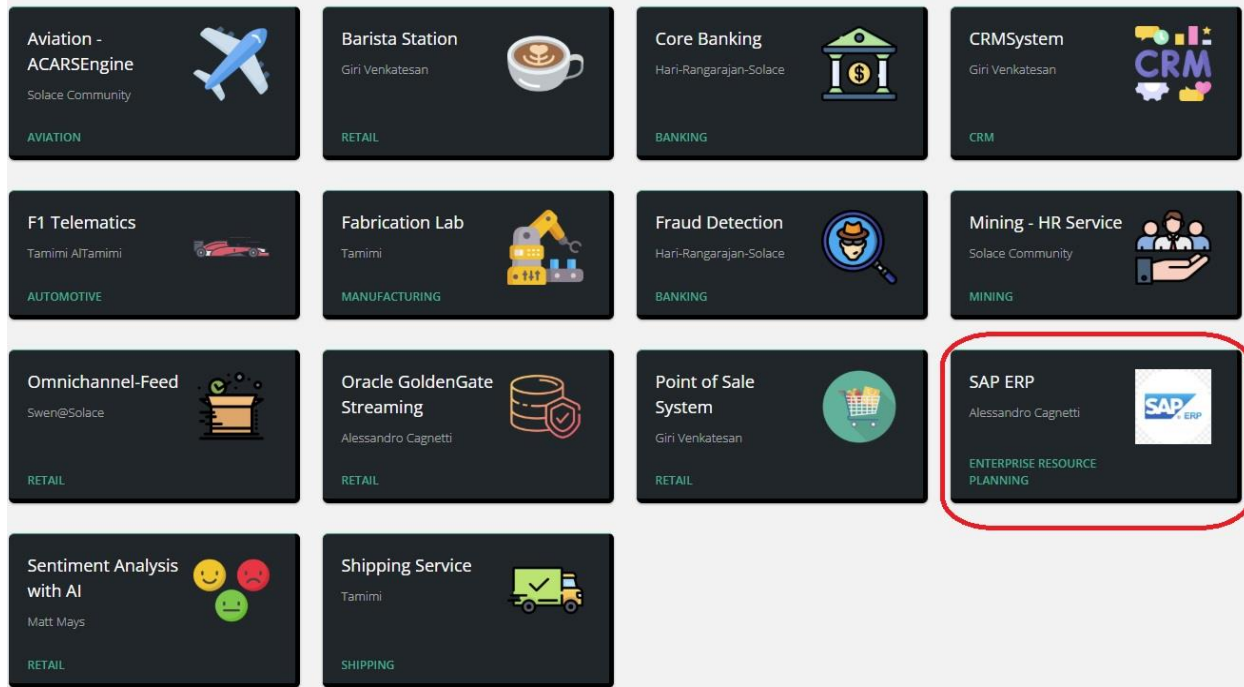
The use of Smart Topics allows you to filter/stream exactly the information you need and the filtering is handled completely by the AEM service so that filtering is not necessary on the client. This makes the construction of the Topic Taxonomy extremely important.

*** This highlighted section is for information only***

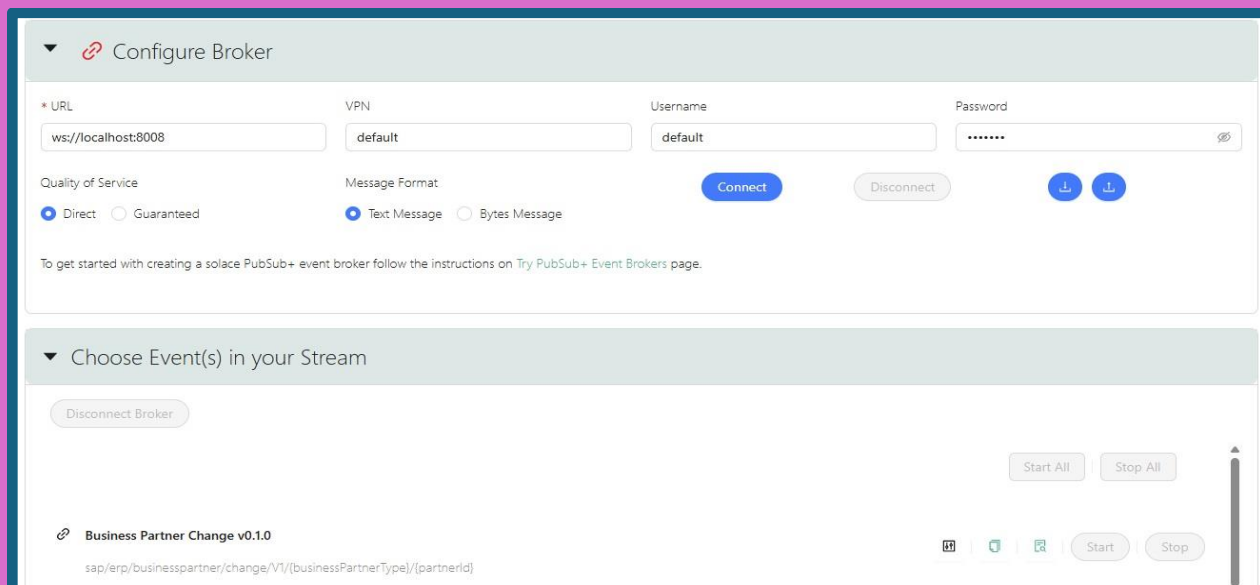
If you would like to use the event generator for your own testing, the instructions are detailed below.

Head to <http://feeds.solace.dev>

Community Feeds

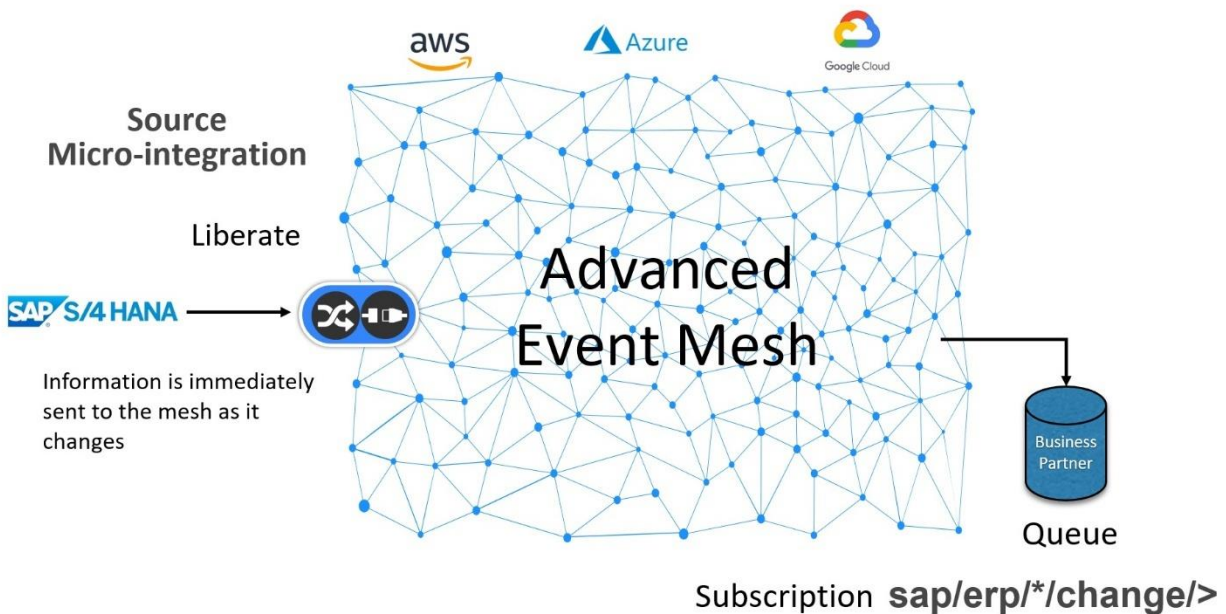


You will notice that there are many feeds and you can experiment with these as you wish. We are using the SAP ERP Feed for this workshop. For the connection details, use the broker connection details for the WebMessaging protocol. Once you are connected, you can then select the Events you would like to test.



3 AEM and Queues

Queues can be used by customers to ensure guaranteed delivery of messages in circumstances where delivery of messages might not be possible. For example, what if the target system has been shutdown for maintenance. For example, perhaps you have an application that needs to be updated with the new business partner record but it has been stopped for maintenance. Without taking the necessary steps, records could potentially be lost. This is one situation where Queues can be really helpful.



A Queue will be created and uses the same Subscription concept to pull the messages into a staging area and will persist them safely in order until the subscribing application is available to receive them. In cases like this, clients are consuming the events from the queue and not consuming based on Topics. This is an important concept and will be illustrated below as a way to guarantee delivery.

4 Setting the stage - Business case scenario #1

Objective:

This hands-on workshop will guide you through a series of exercises designed to introduce the pattern of modern integration using the “Micro Integration” concept. The concepts presented here can be used by existing Integration Suite, Cloud Integration customers to facilitate and compliment their approach to integration.

We'll begin with a discussion around conventional integration design using SAP Integration Suite (iFlows), then illustrate how the EDA micro integration pattern can be used to bring an enhanced level of flexibility and agility to all future integration scenarios.

We believe you will quickly discover and understand why **event-driven architecture (EDA)** can be used with Cloud Integration to deliver better agility, scalability, and decoupling.

Business Case Scenario:

Business Requirement:

Synchronize any updates made to a Business Partner record in SAP S/4HANA with Salesforce (SFDC).

How Would You Approach This?

Traditional Approach using an iFlow



** It should be noted that this is a simplified iFlow that does not include error handling or guaranteed delivery of the messages which would typically include a queuing mechanism.**

Implementation Path:

1. Use **SAP Integration Suite** to create an **iFlow**.
2. Connect to **SAP S/4HANA** via OData or IDoc adapter to fetch supplier updates.
3. Perform any necessary transformations.
4. Connect to **Salesforce** using the Salesforce adapter or REST API.
5. Push the updated record to Salesforce.

Challenges:

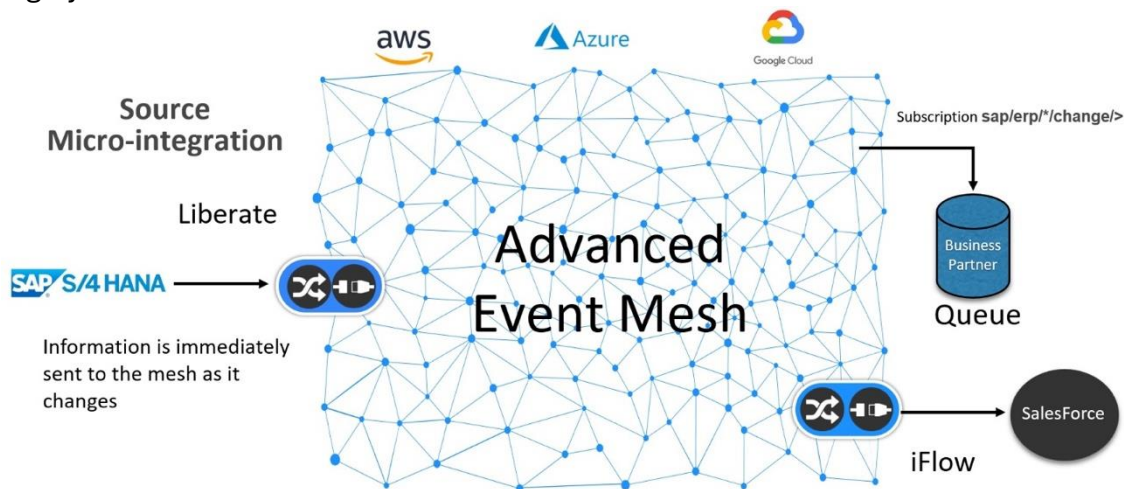
- Receiver is pre-determined when building the iFlow
- Retry logic will be created manually.
- Difficult to extend when new consumers (e.g., Coupa, DocuSign, ServiceNow, data lake, compliance app etc.) need the same event.
- If used, polling mechanisms can create latency and load on the backend.
- If used, scheduled jobs will not deliver the information to the backend in a timely manner leading to out of sync 3rd party systems.

S4 Hana natively produces events and there are several mechanisms that permit S/4 and ECC to publish full data events. So, any changes to the business partner can directly be published on Advanced Event Mesh (AEM) as an event.

We still build an iFlow, but this time it will be like a micro-integration (micro-service) that reacts to an event in real-time. S4 is publishing the data to the Advanced Event Mesh and there is not pre-conceived notion about where that data will go. As a result, S4 Hana and Salesforce systems are now **decoupled**. If one system is down, the other is not affected. You also get **built-in shock absorber** capability of AEM where a sudden deluge of data from S4 Hana will not overwhelm your iFlow. The extra volume of data will simply be buffered on a queue and AEM will trickle feed records to the iFlow at the speed it can be processed.

Modern Approach – Event-Driven Architecture (EDA)

With an EDA Micro Integrations approach, the Business Partner changes will be published to a topic on the Advanced Event mesh. A Queue will be created to subscribe to the relevant events that you want. The iFlow will subscribe to that Queue, receive all relevant changes and react to those events. You now have a Loosely coupled, highly flexible and highly scalable architecture.



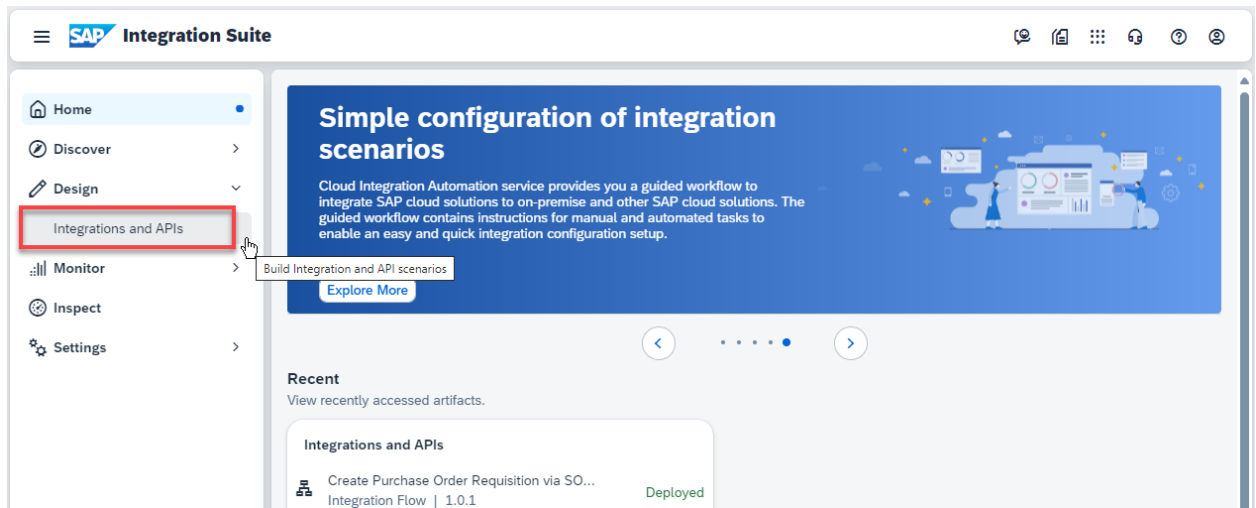
Advantages:

- Loose coupling of systems
- Real-time processing without polling
- Better observability and error isolation

5 Set up packages within Integration Suite, cloud integration

5.1 Access the Cloud Integration service

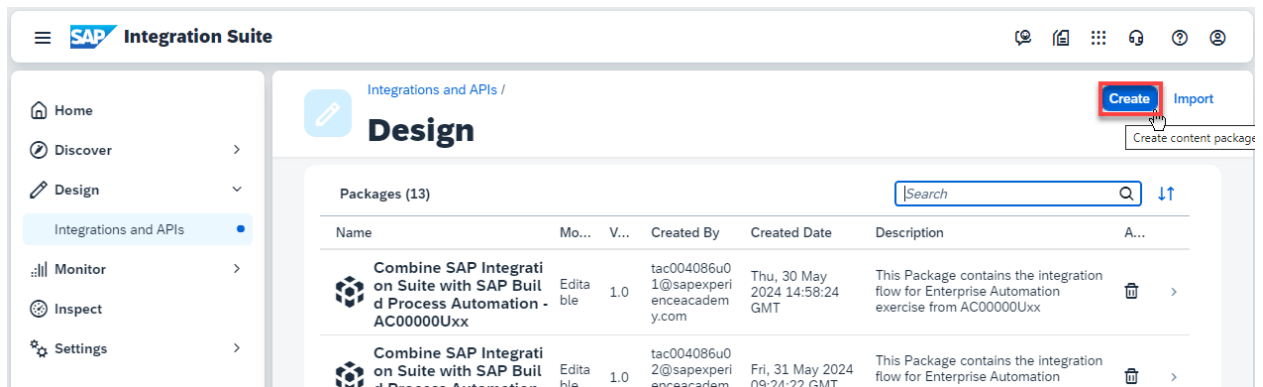
5.1.1 Access the [SAP Integration Suite](#) service and click on Design section in the left-hand menu. Now click on “Integration and APIs” under Design -



5.1.2 Create the integration package

After clicking on “Integration and APIs” you will see a “Design” window on the right-hand side of the screen. This is where you will create your own package.

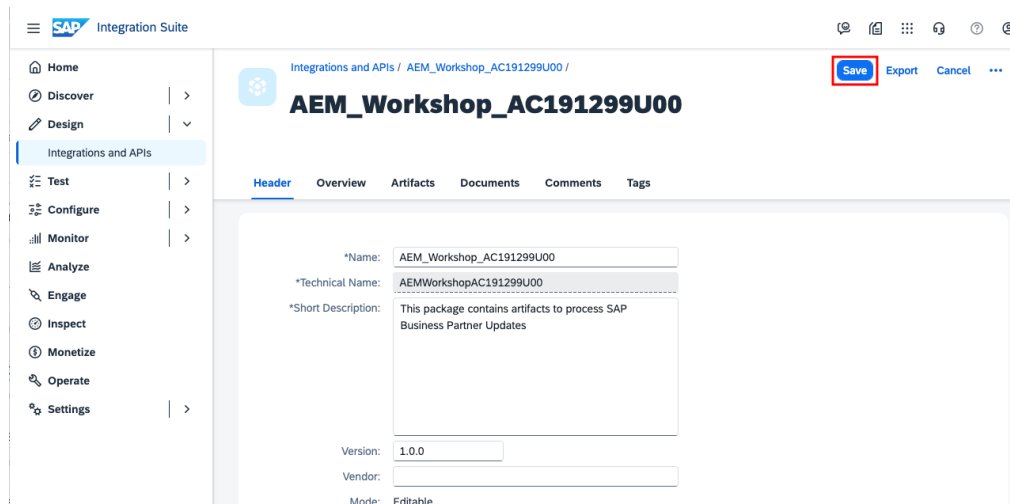
5.1.3 Click on Create to create an integration package.



5.1.4 Provide the following details:

1. Name: AEM_Workshop_AC373526**Unn** where nn represents the last 2 digits of your userid
2. Technical Name will be auto-filled with AEM_Workshop_AC269416**Unn**
3. Short description: This package contains artifacts to process SAP Business Partner Updates.

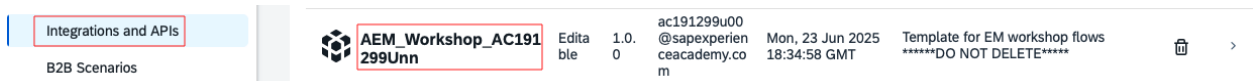
5.1.5 Click on Save once finished.



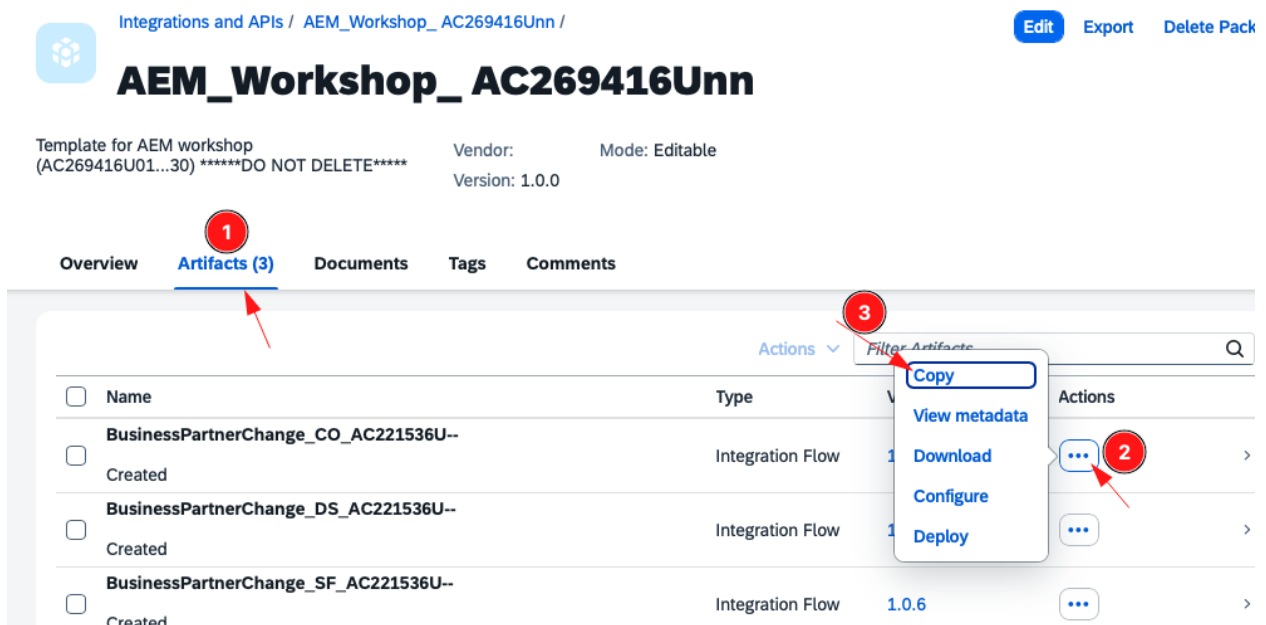
Within an integration package, you can add several artifacts: REST APIs, OData APIs, integration flows, mappings, and more. In our case, we will import templates for integration flows that will be used in the workshop.

5.1.6 You will now import the iFlow templates into your package from the templates package

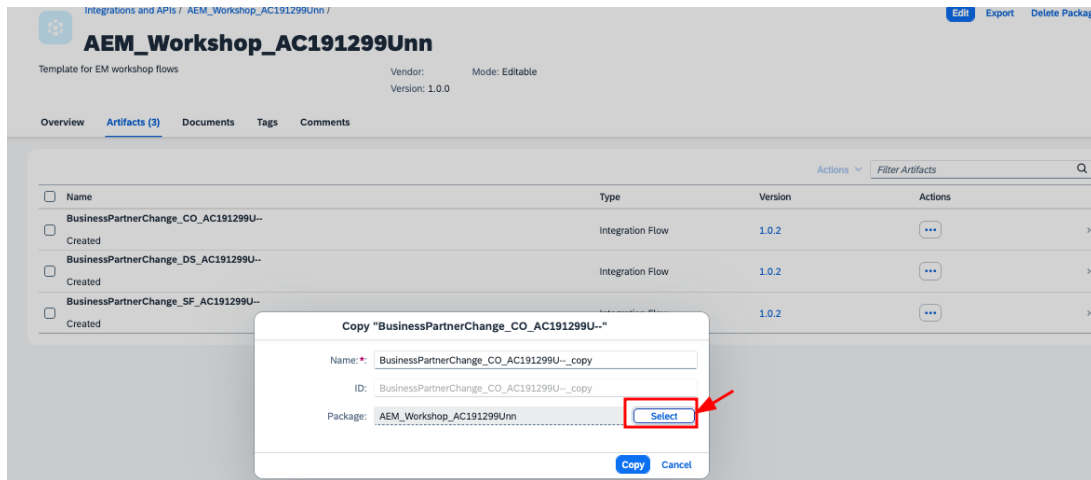
5.1.6.1 Click on Integrations and APIs -> Click on AEM_Workshop_AC373526Unn



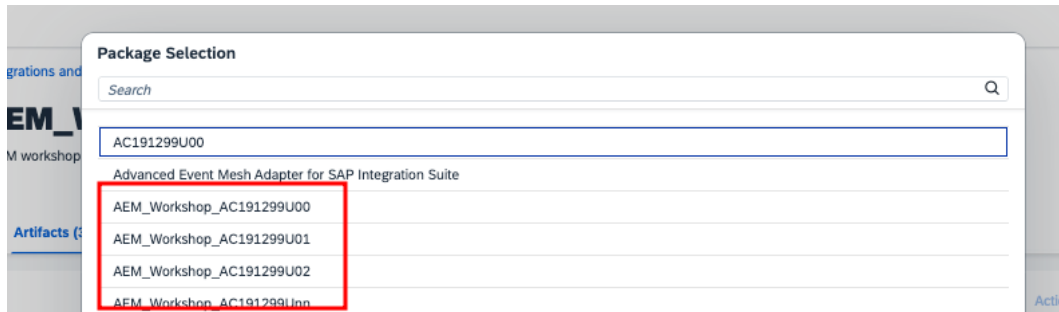
5.1.6.2 Click on Artifacts -> Select one of the template iFlows (by clicking on the 3 dots)



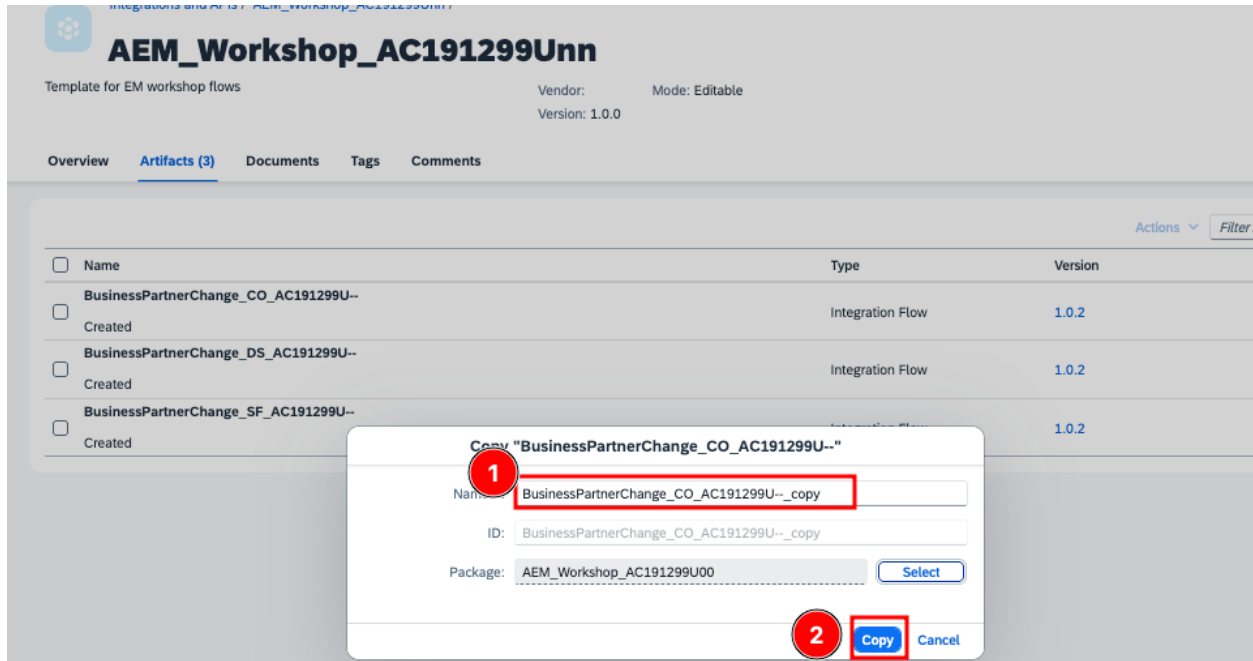
5.1.6.3 Select the target package (your package e.g. AEM_Workshop_AC373526Unn) by clicking on “Select” see screenshot below:



5.1.6.4 Select your package; see screenshot below



- 5.1.6.5 Now type in the name of your iFlow in the Name field. For example, if you selected the BusinessPartnerChange_CO_AC373526U-- iFlow, you would type in BusinessPartnerChange_CO_AC373526U55; if 55 represents the last 2 digits of your userid.



- 5.1.7 Repeat the process for the other 2 template iFlows –

BusinessPartnerChange_CO_AC373526U--

BusinessPartnerChange_CO_AC373526U—

5.2 You should now have 3 iFlows in your Integration Package. Let's check.

Click on “Integrations and APIs” -> Click on AEM_Workshop_AC373526U{your 2 digits} ->

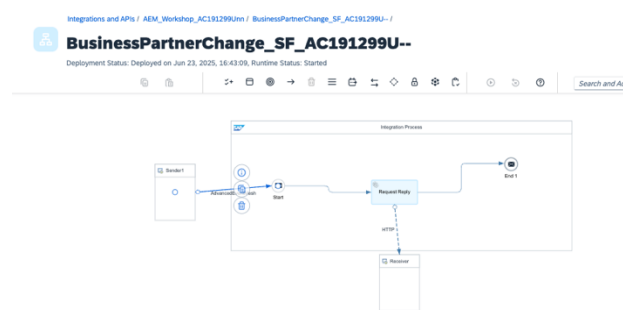
6 Scenario 1 – Publish Business Partner Changed events to Salesforce

In this exercise, you will *subscribe* to Business Partner Changed events published by SAP. To achieve this, define a topic on your AEM broker that will receive the relevant events being published and associate this topic with a queue so that all these events are attracted to that queue. Remember that we previously discussed topic to queue mapping in the 1st warmup exercise.

The iFlow you previously downloaded has most of the configuration already in place (so **you won't have much typing to do!**). The iFlow uses the **SAP AEM Adapter** to read events from the queue on your AEM instance. Within the iFlow, an outgoing call will be made to Salesforce. We will show you the message getting to the Salesforce system and the successful response.

6.1 Overview of iFlow

This iFlow receives Business Partner Change and Create events being published to the

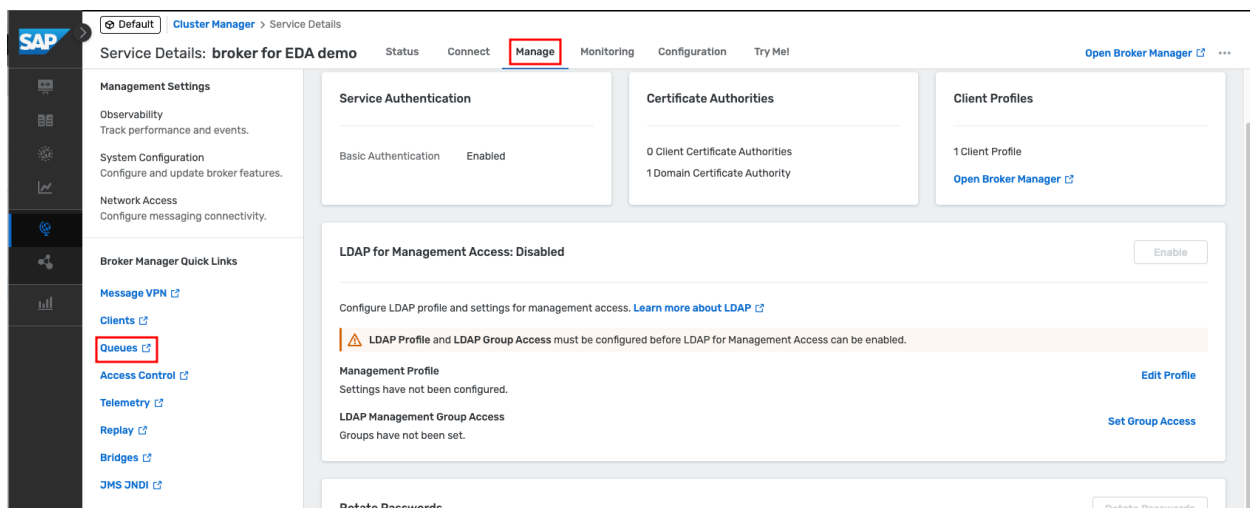


topics shown below. It processes the events by making a HTTP call to the Salesforce application (**a mock service**). In this exercise, each of you will be publishing change events (and some of you are using the same broker).

sap/erp/businesspartner/change/V1/>
sap/erp/businesspartner/create/V1/>

6.2 Create required queues and subscriptions for Salesforce iFlow

Go to [Advanced Event Mesh Console](#) -> **Cluster Manager** -> {your service} -> **Manage** -> **Queues** - to open the Broker UI



Click on the "+ Queue" button to bring up the create queue dialog.

The screenshot shows the SAP Queue management interface. The left sidebar contains the navigation menu with the following items: broker for EDA demo, broker-for-eda-demo, Change VPN, Messaging, Message VPN, Clients, Queues (selected), Connector Wizards, Access Control, Telemetry, Replay, Bridges, JMS JNDI, and Try Me!.

The main area displays a table of queues. The table has the following columns: Queue Name, Incoming, Outgoing, Access Type, Partition Count, Messages Queued (%), Messages Queued (msgs), Messages Queued (MB), Messages Queued Quota (MB), Consumers, Replay State, and Durable. The table contains two rows of data:

Queue Name	Incoming	Outgoing	Access Type	Partition Count	Messages Queued (%)	Messages Queued (msgs)	Messages Queued (MB)	Messages Queued Quota (MB)	Consumers	Replay State	Durable
AC191299U00_SF_IN	On	On	Exclusive	0		2	0.0010	5,000	0	N/A	Yes
AC191299U00_SF_OUT	On	On	Exclusive	0		0	0	5,000	0	N/A	Yes

A red box highlights the "+ Queue" button in the top right corner of the interface.

Provide the name as given (in the next sections).

The screenshot shows the 'Create Queue' dialog box. It has a title bar 'Create Queue' and a 'Queue Name' input field. Below the input field are two buttons: 'Cancel' and 'Create'.

Create the following queue and provide the details as given (copy & paste where appropriate).

1. AC373526Unn_SF_IN queue

- Name: AC373526Unn_SF_IN
- Owner: solace-cloud-client

Once the queue is created, click on the queue name in the list, navigate to the Subscriptions tab and open the subscriptions dialog.

- For AC373526Unn_SF_IN, add the subscriptions:

sap/erp/businesspartner/change/V1/>

sap/erp/businesspartner/create/V1/>

6.3 Obtain AEM Broker Connection Credentials

Before heading back to Integration Suite, let's head to our [Advanced Event Mesh Console](#) and go to **Cluster Manager** -> **{your service}**. Select the connection point and protocol that you want to use to connect your Integration Suite flows by going to the "Connect" tab, order by protocol, then click on Solace Messaging. Make a note of the connectivity details underneath "Solace Messaging" (click on the section to open it up). We will need these details in the next steps when configuring your iFlows.

The screenshot shows the SAP Integration Suite console interface. The 'Connect' tab is active, displaying a table of supported client libraries for Solace Messaging. The 'Solace Java API' library is highlighted. The 'Connection Details' panel on the right shows the following information:

Field	Value
Username	solace-cloud-client
Password	*****
Message VPN	broker-for-eda-demo
Host URIs	Secured SMF URI https://mr-connection-sb2zodlaa7b.messaging.solace.cloud:55443

The connect tab lists all the various connectivity details for the various supported protocols. The SAP Integration AEM adapter uses Solace Messaging Format, which is AEM's very own protocol with a broad feature set support. [Solace Message Format](#) (SMF) is the underlying messaging protocol for SAP Integration Suite, advanced event mesh.

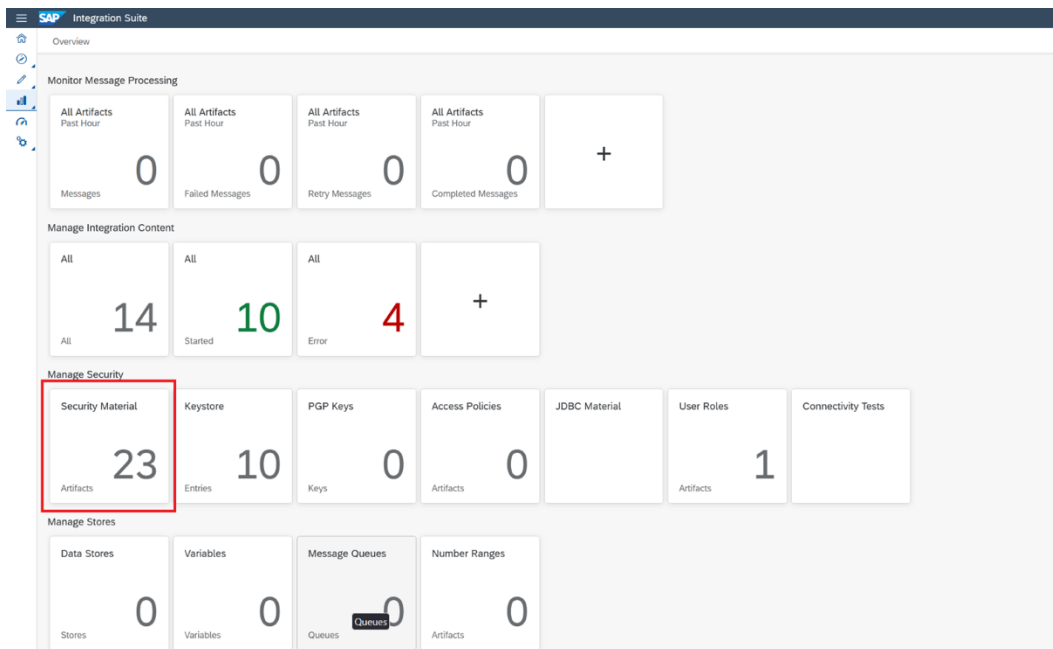
Each AEM service also comes with a default client user called solace-cloud-client that is configured for convenience reasons and is allowed to publish and subscribe to all topics. We will be using this user for all our iFlows.

6.4 Security Configuration

N.B. You only need to do this section (6.4) if you are bringing your own broker rather than the one provided by the workshop. All this has already been done for those using the workshop provided broker.

Rather than entering the AEM instance password directly on the iFlow and making it visible to everyone with access to the iFlow, we will create a **SecureParameter** which will store the password securely and we then just reference this in our iFlows.

Go to [Integration Suite](#) -> **Monitor** -> **Manage Security** -> **Security Material**.



In here, create security credentials for your AEM broker service.

Create **SecureParameter** AC373526UnnBrokerUserPass and store the password for your solace-cloud-client application user credentials; nn represents the last 2 digits of your userid.

6.5 Configure and Deploy

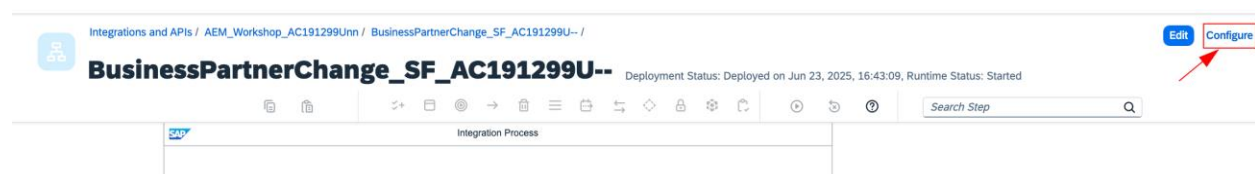
BusinessPartnerChange_SF_AC373526Unn

Navigate to the iFlow – Go to [Integration Suite](#) -> **Design** -> **Integrations and APIs** -> **AEM_Workshop_AC373526Unn** -> **Artifacts** -> **BusinessPartnerChange_SF_AC373526Unn**

6.5.1 Configure AEM Adapter connection credentials

N.B. You only need to do this section (6.5.1) if you are bringing your own broker rather than the one provided by the workshop. All the pre-populated values for the iFlow should be correct for those using the workshop provided broker.

Click on the configure button:



Click on the Sender tab and populate the connection credentials.

- Type in the AEM connection url you obtained from 6.3 above
- Type in the message vpn you obtained from 6.3 above
- Type in the Password Secure Alias you created from 6.4 above
- Click on Save
- Click on Close

Configure "BusinessPartnerChange_SF_AC191299U--"

[Sender](#)

Sender: Sender1

Adapter Type: AdvancedEventMesh

Connection

Host: tcps://mr-connection-3f3ava6d2q.messaging.solace.cloud:55443

Message VPN: us_nsqaem_workshop

Username: solace-cloud-client

Password Secure Alias: AEM_NSQA

Save Deploy Close

6.5.2 Edit the iFlow and add your queue name to the AEM Adapter configuration; **click on the AEM adapter arrow.**

- 1) Hit the Edit Button so that you can change the iflow
- 2) Click the "Dashed Line" that represents the connection to the receiver, which in this case is the AEM service.
- 3) Specify the Queue name where your messages are being stored.

Integrations and APIs / AEM_Workshop_AC191299U00 / BusinessPartnerChange_SF_AC191299U00 /

BusinessPartnerChange_SF_AC191299U00

Deployment Status: Deployed on Jun 23, 2025, 16:43:09, Runtime Status: Started

Search Step

AdvancedEventMesh

General Connection Processing

SENDER PROCESSING DETAILS

Consumer Mode: Guaranteed

Run on a single worker node?: ☒

Parallel Consumers: 1

Queue Name: AC191299U00_SF_in

6.5.3 Additional prefix to URL

To make sure your HTTP call gets to the correct destination (and to help identify individual requests in case of troubleshooting), add the following suffix to the URL for the HTTP call

/sf/ AC373526U**nn**(Replace “nn” with your userID)

'sf' is the identifier for the salesforce mock service

Integrations and APIs / AEM_Workshop_AC191299U00 / BusinessPartnerChange_SF_AC191299U00 /

BusinessPartnerChange_SF_AC191299U00

Deployment Status: Deployed on Jul 24, 2025, 09:52:00, Runtime Status: Started

Click on Edit

Search Step

Click on the arrow so that the HTTP properties become visible

HTTP

General Connection

CONNECTION DETAILS

Address: https://aem-workshop-us2.mockoon.app/sf/AC191299U00

Query:

3 Add the suffix /sf/AC191299U00 to the URL as shown

6.5.4 Click on Save at the top right of the screen

Integrations and APIs / AEM_Workshop_AC191299U00 / BusinessPartnerChange_SF_AC191299U00 /

BusinessPartnerChange_SF_AC191299U00

Deployment Status: Deployed on Jun 19, 2025, 00:35:15, Runtime Status: Started

Save Save as version Deploy Cancel Delete

6.5.5 Deploy the iFlow

Integrations and APIs / AEM_Workshop_AC191299U00 / BusinessPartnerChange_SF_AC191299U00 /

BusinessPartnerChange_SF_AC191299U00

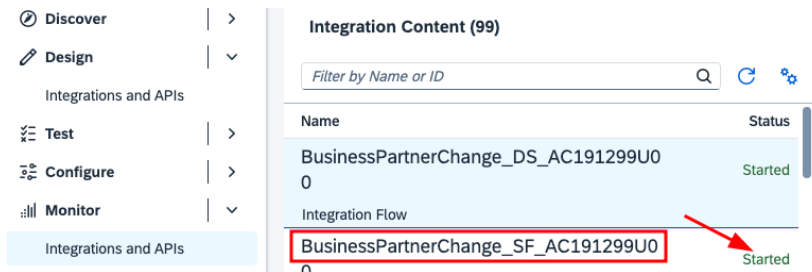
Deployment Status: Deployed on Jun 19, 2025, 00:35:15, Runtime Status: Started

Save Save as version Deploy

6.5.6 Check on the Deployment progress

Make sure the iFlow goes to a Started state.

Go to [Integration Suite](#) -> **Monitor** -> **Integrations and APIs** -> **Manage Integration Content**



6.6 Test the BusinessPartnerChange_SF_AC373526Unn iFlow

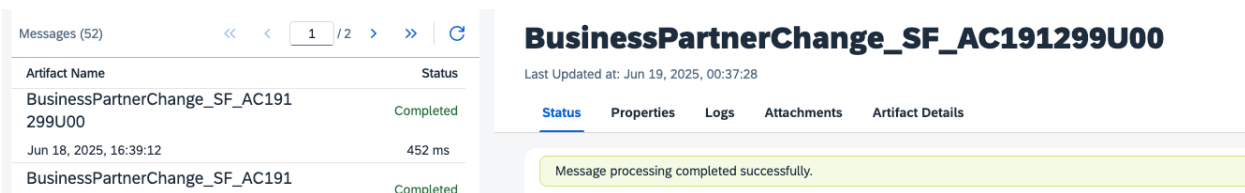
Once everybody is ready, we will start the feeds again and events will start to flow to all your iFlows

We will show you your message being received by the Salesforce mock service.

You should also familiarise yourself with the error logs for when things don't go quite so well!

Go to [Integration Suite](#) -> **Monitor** -> **Integrations and APIs** -> **Monitor Message Processing**

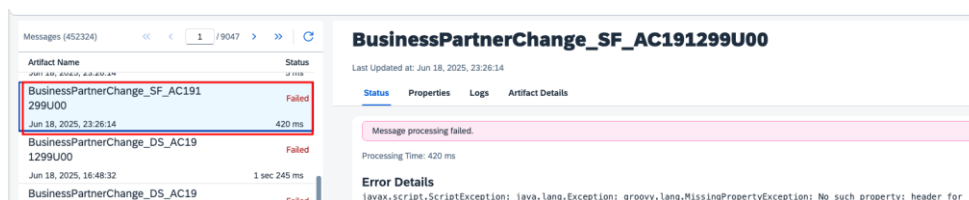
For a successful response, you should see something like:



If you don't get a response, check on the error logs for the iFlow in Integration Suite

Go to [Integration Suite](#) -> **Monitor** -> **Integrations and APIs** -> **Monitor Message Processing**

Have a look at the logs and troubleshoot the error.



7 Setting the stage - Business case scenario #2

Business Case Scenario:

Business Requirement:

*In addition to updating Salesforce, the business now requires that any changes made to a Business Partner (Supplier) record in SAP S/4HANA are also sent to **Coupa**, to ensure supplier profiles are accurately reflected in procurement workflows.*

How Would You Approach This Today?

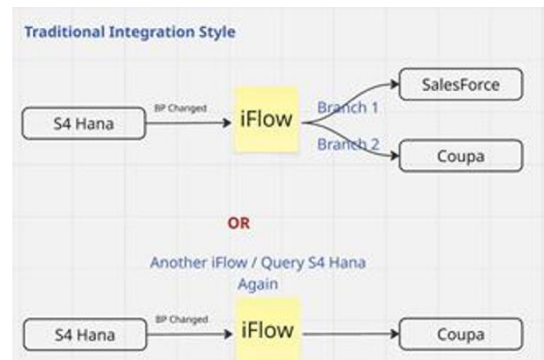
Traditional Approach – Point-to-Point (P2P) Integration

Implementation Path:

1. You may extend your existing iFlow to also call **Coupa's Supplier API**, or
2. Build a **second iFlow** solely for Coupa

Challenges Amplified:

- Increased payload transformation complexity
- Error handling across systems becomes harder to manage
- Deployments take longer with every new consumer
- Adding a third or fourth downstream system (e.g., Data Lake, SAP Analytics Cloud, etc.) would require further duplication or chaining

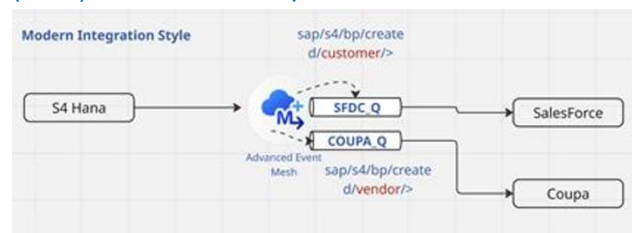


Again, we are not going to use the P2P approach today, but we will use event-driven integration approach.

Modern Approach – Event-Driven Architecture (EDA) – Seamless Expansion

With SAP BTP Advanced Event Mesh (AEM):

1. No changes are needed in the source system (S/4HANA continues publishing Business Partner events).
2. Coupa integration becomes just another subscriber.
3. You simply build a second event-consumer iFlow in Integration Suite for Coupa.



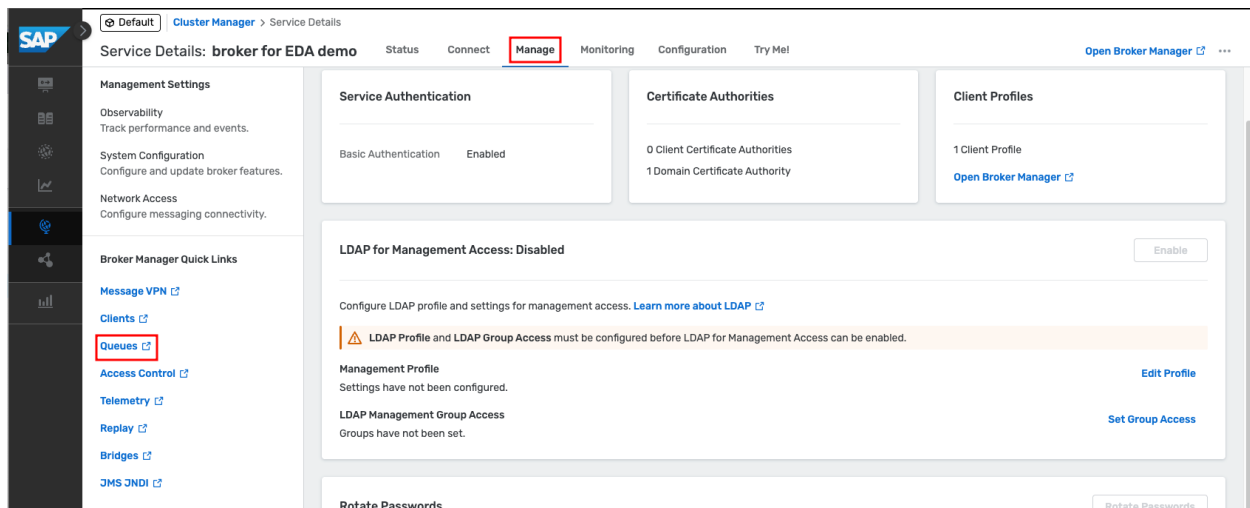
Benefits Realized:

1. Business logic for each consumer is encapsulated independently
2. No regression risk to Salesforce flow
3. Future systems (e.g., Compliance Monitoring, Risk Engines) can subscribe instantly without any changes to the event producer

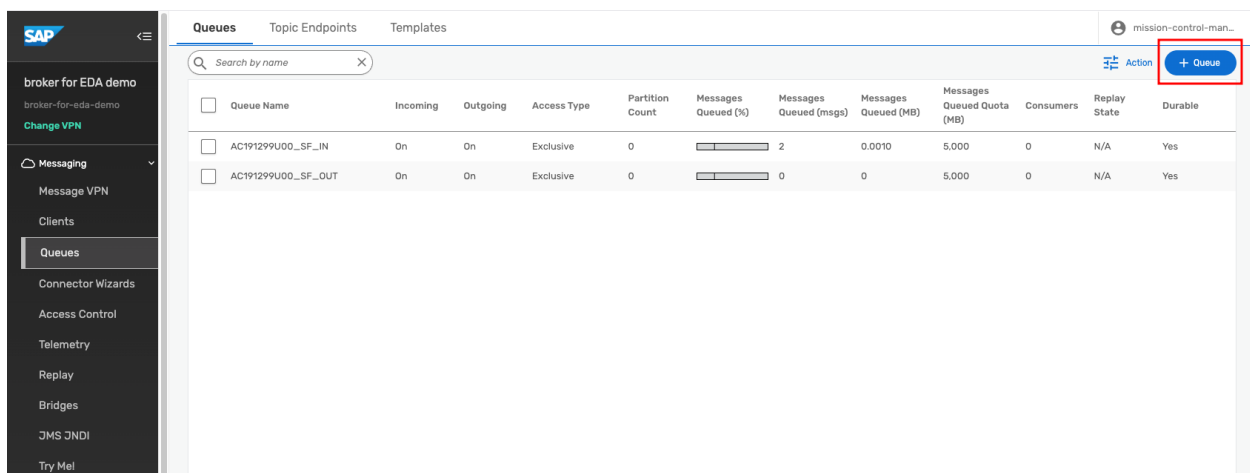
8 Scenario 2 – Publish Business Partner Changed events to Coupa

8.1 Create required queues and subscriptions for Coupa iFlow

Go to [Advanced Event Mesh Console](#) -> **Cluster Manager** -> {your service} -> **Manage** -> **Queues** - to open the Broker UI



Click on the "+ Queue" button to bring up the create queue dialog.



Create the following queue and provide the details as given (copy & paste where appropriate).

1. AC373526Unn_CO_IN queue

- Name: AC373526Unn_CO_IN
- Owner: solace-cloud-client

Once the queue is created, click on the queue name in the list, navigate to the Subscriptions tab and open the subscriptions dialog.

- For AC373526Unn_CO_IN, add the subscriptions:

sap/erp/businesspartner/change/V1/>

sap/erp/businesspartner/create/V1/>

8.2 Configure and Deploy

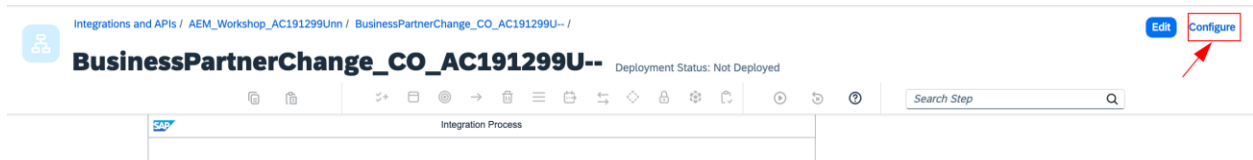
BusinessPartnerChange_CO_AC373526Unn

Navigate to the iFlow – Go to [Integration Suite](#) -> **Design** -> **Integrations and APIs** -> **AEM_Workshop_AC373526Unn** -> **Artifacts** -> **BusinessPartnerChange_CO_AC373526Unn**

8.2.1 Configure AEM Adapter connection credentials

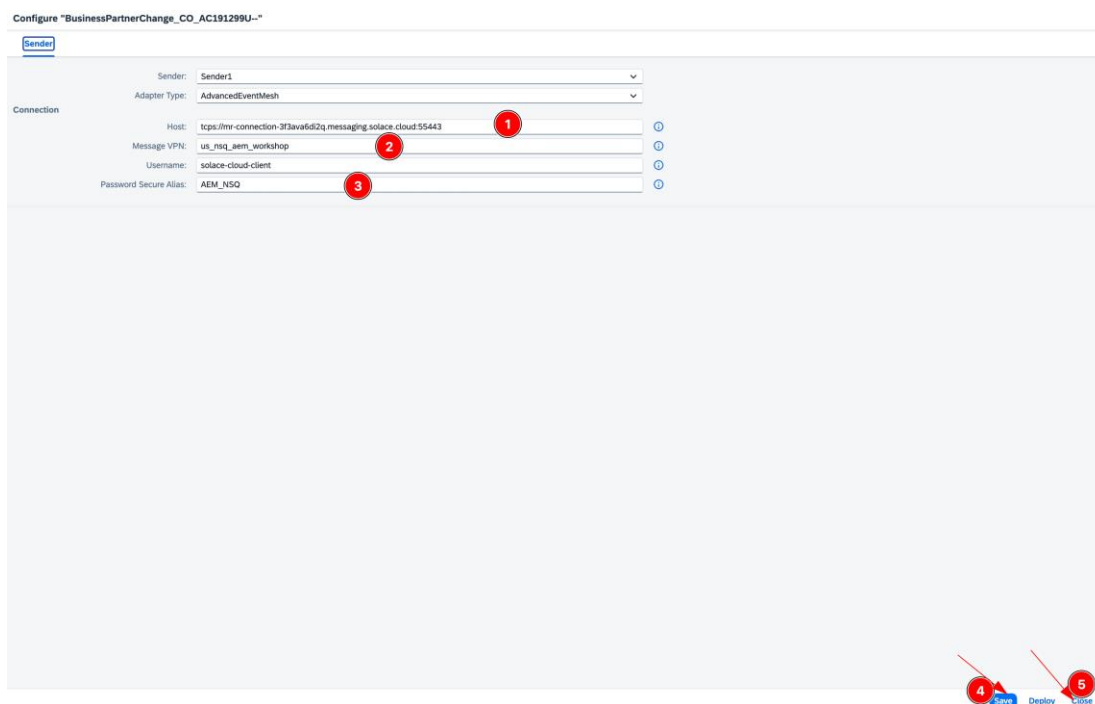
N.B. You only need to do this section (8.2.1) if you are bringing your own broker rather than the one provided by the workshop. All the pre-populated values for the iFlow should be correct for those using the workshop provided broker.

Click on the configure button:



Click on the Sender tab and populate the connection credentials.

- Type in the AEM connection url you obtained from 6.3 above
- Type in the message vpn you obtained from 6.3 above
- Type in the Password Secure Alias you created from 6.4 above
- Click on Save
- Click on Close



8.2.2 Edit the iFlow and add your queue name to the AEM Adapter configuration; **click on the AEM adapter arrow**

Integrations and APIs / AEM_Workshop_AC191299U00 / BusinessPartnerChange_CO_AC191299U00 /

BusinessPartnerChange_CO_AC191299U00 Deployment Status: Not Deployed

Search Step

Integration Process

Sender1

AdvancedEventMesh

Start

Request Reply

End 1

HTTP

Receiver

AdvancedEventMesh

General Connection Processing

SENDER PROCESSING DETAILS

Consumer Mode: Guaranteed

Run on a single worker node?: ☒

Parallel Consumers: 1

Queue Name: AC191299U00_CO_IN

Selector:

8.2.3 Additional prefix to URL

To make sure your HTTP call gets to the correct destination (and to help identify individual requests in case of troubleshooting), add the following suffix to the URL for the HTTP call

/co/ AC373526U00(Replace “nn” with your userID)
'co' is the identifier for the Coupa mock service

Integrations and APIs / AEM_Workshop_AC191299U00 / BusinessPartnerChange_CO_AC191299U00 /

BusinessPartnerChange_CO_AC191299U00 Deployment Status: Deployed on Jul 24, 2025, 10:01:45, Runtime Status: Started

Click on Edit

Search Step

Integration Process

Sender1

AdvancedEventMesh

Start

Request Reply

End 1

HTTP

Receiver

HTTP

General Connection

CONNECTION DETAILS

Address: https://sap-aem-wshop.us2.mockoon.app/co/AC191299U00

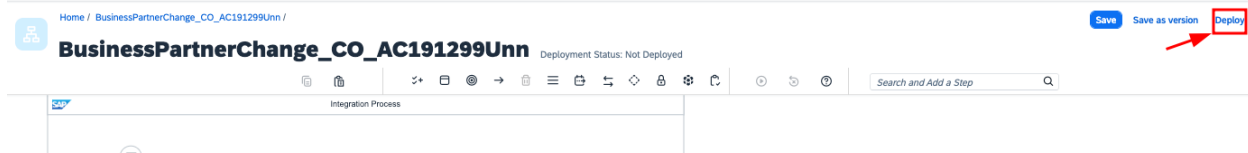
Click on arrow so that the HTTP properties become visible

Add the suffix /co/AC191299U00 to the URL as shown

8.2.4 Click on Save at the top right of the screen



8.2.5 Deploy the Coupa iFlow



8.2.6 Check on the Deployment progress of the Coupa iFlow

Make sure the Coupa iFlow goes to a Started state.

Go to [Integration Suite](#) -> **Monitor** -> **Integrations and APIs** -> **Manage Integration Content**

8.3 Test the BusinessPartnerChange_CO_AC373526Unn iFlow

Once everybody is ready, we will start the feeds again and events will start to flow to all your iFlows

We will show you your message being received by the Coupa mock service.

Notice that your salesforce iFlow will also have processed the message. Both Salesforce and Coupa are subscribing to the same topic and therefore receive the same data that is published.

9 Demos – Integrations beyond iFlows: If Time permits

10 Optional: Understanding Retry, Error handling, unblocking message processing – If time permits

This exercise will demonstrate the benefits you get by adopting a publish/subscribe methodology even if you only have 1 downstream subscriber.

In this scenario, the downstream application becomes unavailable (we will arrange this). You will make some configuration changes to your queues to implement error/retry-handling and ensure that undeliverable messages do not block processing. All this without writing any code!

10.1 Adapter settings

Let's take a look at some of the relevant AEM adapter settings that control this behaviour.

The screenshot displays the SAP Integration Suite interface for the integration process **BusinessPartnerChange_SF_AC191299U--**. The deployment status is "Deployed on Jun 23, 2025, 16:43:09, Runtime Status: Started". The integration process diagram shows a flow from **Sender1** through an **AdvancedEventMesh** adapter to a **Request Reply** component, which then connects to a **Receiver** via an **HTTP** connection, finally reaching **End 1**.

The **AdvancedEventMesh** adapter configuration is shown in the "Processing" tab. The "SENDER PROCESSING DETAILS" section includes the following settings:

- Consumer Mode: **Guaranteed**
- Run on a single worker node?: ☒
- Parallel Consumers: **1**
- Queue Name: **AC191299U00_SF_IN**
- Selector: (empty)
- Acknowledgment Mode: **Automatic On Exchange Complete**
- Settlement Outcome After Maximum Attempts: **Failed**
- Maximum Message Processing Attempts: **2**
- Retry Interval (in ms): **500**
- Maximum Retry Interval (in ms): **500**
- Exponential Backoff Multiplier: **1.0**

Let's look at these settings one by one:

1. Acknowledgement Mode: "Automatic on Exchange Complete"

The most important setting when it comes to not accidentally acknowledging and therefore removing a message from the broker's queue. This setting tells the flow/AEM adapter to only acknowledge (ack) the message after the flow has successfully completed processing the

message. If any error in the processing occurs, the AEM adapter will instead send a negative acknowledgment back (nack) to tell the broker to keep the message and retry it, because it couldn't be successfully processed by the flow. The alternative is to immediately ack the message when it's received, which will always result in the message being removed from the queue even if the flow fails to successfully process the message. (!!)

2. Settlement Outcome After Maximum Attempts: "Failed"

This setting controls the nack type and behaviour, we have two options here:

a) **Failed**, which will nack the message back to the broker and lets the broker check the retry count of the message to trigger retries based on the queue settings and only sending messages to DMQ when the retry count on the message has exceeded the max retry settings on the queue.

N.B. With the Failed setting, values of the Maximum Redelivery Count on the queue and the max. message processing attempts on the adapter are taken into account.

b) **Rejected**, which will nack the message telling the broker to immediately move the message to DMQ when the AEM adapter settings (Maximum Message Processing Attempts) are exceeded irrespective of queue settings.

N.B. With the Rejected setting, only the value for max. message processing attempts on the adapter is taken into account.

3. Max. Message Processing Attempts: 2

Controls how often we want to retry a message inside the iFlow before we "give up" and pass it back to the broker.

4. Retry interval, Max Retry Interval and Exponential Backoff Multiplier

These are all settings that control how quickly we want to retry and whether we want to incrementally increase our retry delay with each failure. A good retry delay value prevents the iFlow from repeatedly retrying a message within a few milli-seconds and gives some time for transient error situations to clear before we retry.

Note that the error handling and retry settings go hand-in-hand with the DMQ and retry settings on the input queue for this flow (queue retry settings multiply with the internal retry settings in the iFlow, e.g. if the iFlow tries 2 times internally every time we pass it a message and the broker is configured to retry the same message 3 times to the broker, then we might get 8 executions before the message is actually stopped being processed and moved to the DMQ [(1 initial attempt + 3 times retry) * 2 times retry inside the iFlow = 8 processing attempts]):

10.2 Queue configuration for retry handling

10.2.1 Create a Dead Message Queue(DMQ)

Using the instructions from 6.2 above, create a DMQ as follows:

1. AC373526Unn_SF_DMQ queue

- Name: AC373526Unn_SF_DMQ
- Owner: solace-cloud-client

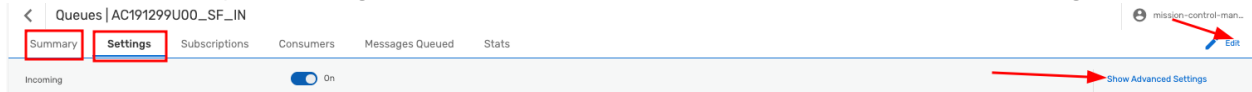
A DMQ is just like any other queue but it is used to hold all messages that have not been successfully processed in a DMQ-enabled iFlow. Not all messages are eligible for dead message processing. There are 4 pre-requisites:

- 1) A DMQ must exist and be associated with the input queue
- 2) The DMQ eligibility must be set by the publisher on the incoming message
- 3) TTL turned on
- 4) Retry settings configured on the AEM adapter.

10.2.2 Amend the Input queue as follows

Go to [Advanced Event Mesh Console](#) -> **Cluster Manager** -> **{your service}** -> **Manage** -> **Queues** - to open the Broker UI -> **Click on the relevant queue** (e.g. AC373526Unn)

Click on Summary or Settings then click on Edit. Next click on Show Advanced Settings.



Add the following configurations

- DMQ Name: AC373526Unn_SF_DMQ
- Redelivery: enabled
- Try Forever: disabled
- Maximum Redelivery Count: 2

A screenshot of the DMQ settings form in the Advanced Event Mesh Console. The 'DMQ Name' field is highlighted with a red box and contains the text 'AC191299U00_SF_DMQ'. Below this, there are several configuration options: 'Respect TTL' (toggle set to 'Off'), 'Maximum TTL (sec)' (input field with '0'), 'Redelivery' (toggle set to 'On'), 'Try Forever' (toggle set to 'Off'), 'Maximum Redelivery Count' (input field with '1'), 'Delayed Redelivery' (toggle set to 'Off'), and 'Multiplier' (input field with '2.00').

Now that all the required retry configuration is in place, you will send a message and observe the retry behaviour. How many retries do you expect when you publish a message on the input queue for the salesforce iFlow and a problem is encountered within the flow?

10.2.3 Publish a message to the input queue for the salesforce iFlow

As mentioned in 10.2.1 above, the **publisher must set DMQ eligibility** when publishing the message if you want a message to be a candidate for DMQ processing and handling. Carefully follow the instructions below on how to activate this setting in Try Me!

Follow the instructions from **Error! Reference source not found.** to access the Try Me! utility to publish to the iFlow.

See the screenshot below and the following instructions on how to access the DMQ settings in Try Me!

Send and Receive

Publisher

Establish Connection
 ❌ Not Connected

Broker URL
 ws://mc-connection-3f3a66d2.messaging.sofiacloud.de

Message VTN
 us_mq_aem_workshop

Client Username
 sofiacloud-client

Client Password

Publish
 Select a topic or queue to publish to

☒ Topic ☐ Queue

[Learn about topic taxonomy](#)

Think Event!
 Build your smart topic to describe an event matching your use case! [Learn more about best practices](#)

Topic Structure
 event / verb / property / property / property

Create a topic using the topic builder

Noun Verb Property Property Property
 flight boarding #1234 yow awr

Publish to topic
 aomeCo/sap/44/sap/customer/created/ACT123456789

Delivery Mode
☒ Direct ☐ Persistent

DMQ Eligible ☒

Message Priority (0 to 9)

Time to Live (sec)

Reply To Topic

Correlation Id

Message Content
 {
 "airline": "ExampleAirline",
 "region": "Ontario",
 "requestId": 443344,
 "flight": {
 "flightModel": "boeing737"
 }
 }

Publish

Messages Published
 0 Direct 0 Persistent

- I. Click on the drop-down arrow to reveal the connection credentials dialogue
- II. Key in the AEM password in the Client Password field
- III. Click on Connect
- IV. Click on show Advanced
- V. Click on the Topic radio button
- VI. Toggle the DMQ eligible button to on
- VII. Paste the message content from 6.6 above in the message content window
- VIII. Click on Publish

Check the processing logs to get more details on the failure. How many retries do you expect to see based on the settings configured on the queue and AEM adapter?

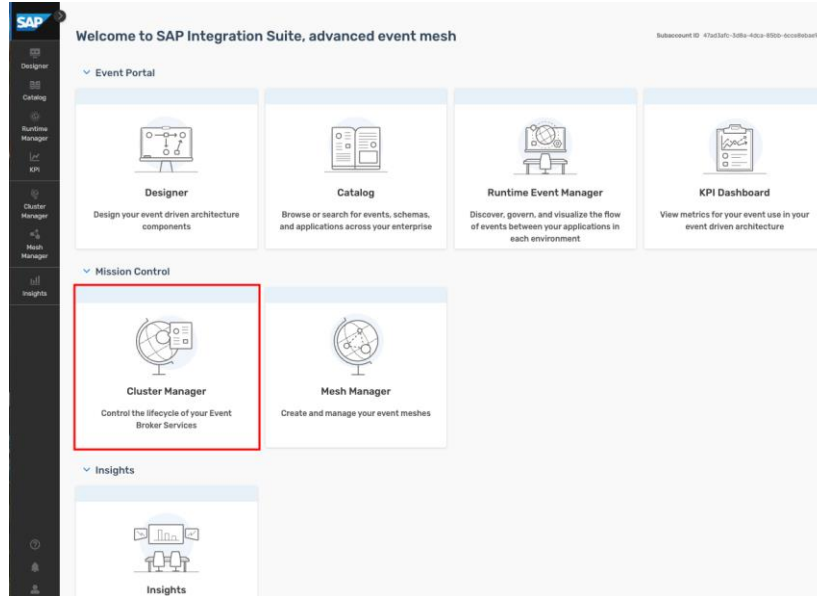
Go to [Integration Suite](#) -> **Monitor** -> **Integrations and APIs** -> **Monitor Message Processing**

11 APPENDIX – CONNECTING TO THE TRY ME UTILITY

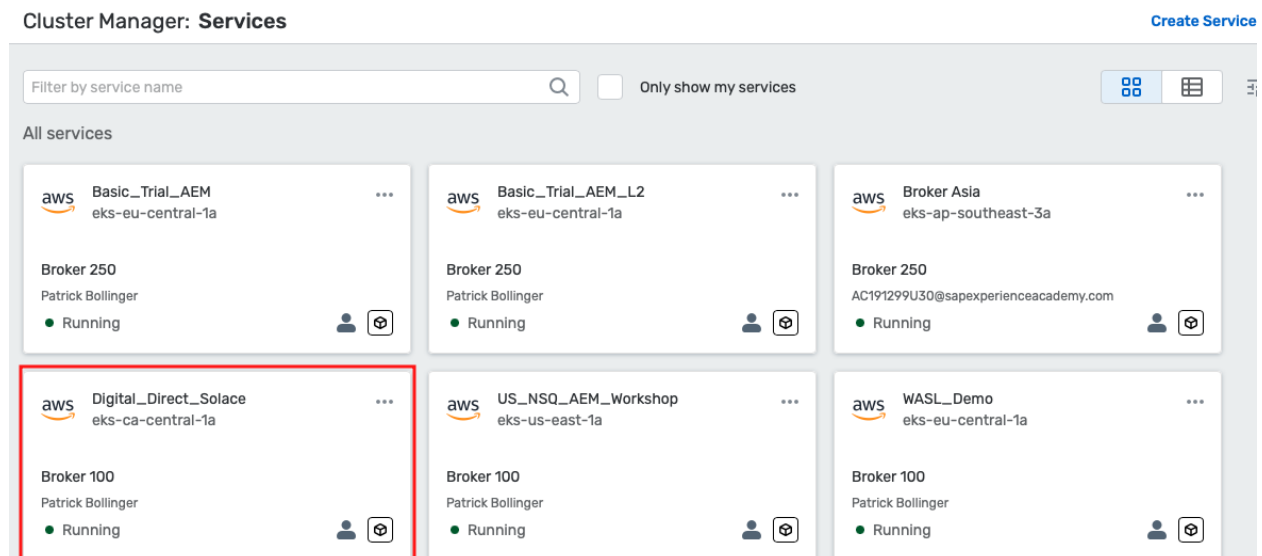
11.1 Obtain connection credentials

The Try Me! utility is a JavaScript application for quickly getting up and running with testing smart topics. The application uses Web Sockets and so you'll need to select the appropriate connection credentials.

Use the link [Advanced Event Mesh](#) and click on **Cluster Manager**



Next, select the broker that is being used for this workshop: **'Digital_Direct_Solace'**



After selecting the AEM instance above,

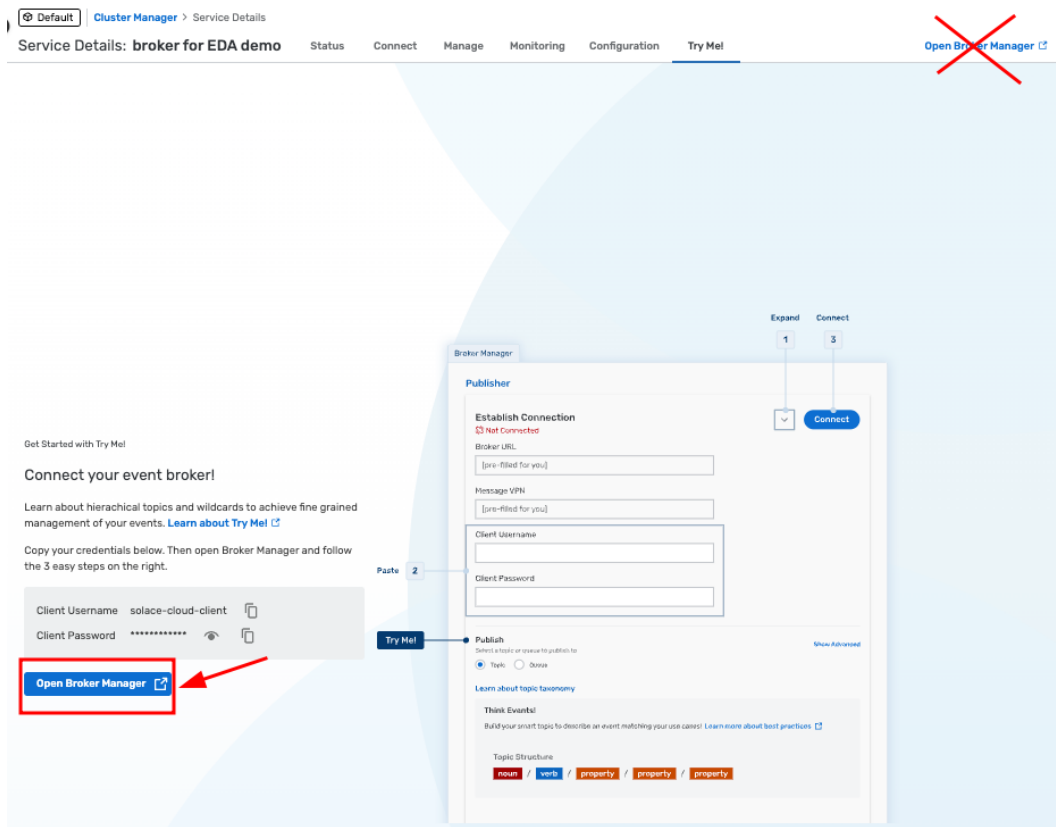
- 1 - click on the "Connect" tab
- 2 - order by protocol
- 3 - click on Solace Web Messaging.

Make a note of the connectivity details underneath "Solace JavaScript API" (click on "Solace JavaScript API" to reveal the connection credentials).

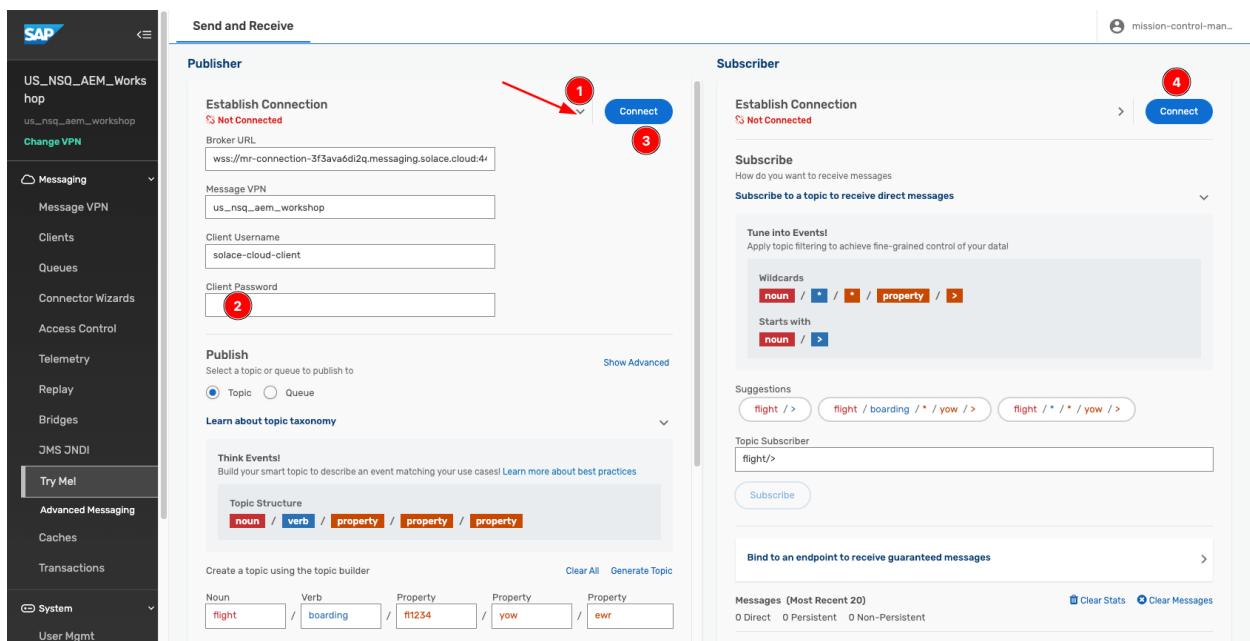
The screenshot shows the SAP Cluster Manager interface. The top navigation bar includes tabs for Default, Cluster Manager, and Service Details. The main content area is titled "Service Details: US_NSQ_AEM_Workshop" and has tabs for Status, Connect, Manage, Monitoring, Configuration, and Try Me. The "Connect" tab is active, showing a "Connect Using a Supported Client Library" section. This section lists various client libraries, including Solace Messaging, Solace Web Messaging, AMQP, MQTT, and REST. The "Solace JavaScript API" is highlighted with a red circle (4). To the right, a modal window titled "Solace JavaScript API" is open, displaying connection details: Username (solace-cloud-client), Password (*****), Message VPI (us_nsq_aem_workshop), and Host URIs (Secured Web Messaging URI: ws://mq-connection-3f5avak2q.messaging.solace.cloud:443). A "Get Started" button is at the bottom of the modal. A red circle (2) is placed over the "Open Broker Manager" link in the top right corner.

You will use these credentials shortly within the Try Me Tab. Select the Try me Tab and proceed with "Open Broker Manager".

This screenshot shows the same SAP Cluster Manager interface, but with the "Try Me" tab highlighted in the top navigation bar. A red arrow points to the "Try Me" tab. The main content area remains the same, showing the "Connect Using a Supported Client Library" section. The "Solace JavaScript API" is still highlighted with a red circle (4). The modal window is no longer open.



Once the Broker Manager is open, select the “Try Me” option from the left side of the menu. You will then use the credentials that you copied above to populate the left side of the screen...AKA the Publisher Side. Once the publisher side says “connected”, you can simply hit the “Connect” button on the right side to also connect your subscriber.



You are now connected to the AEM service with a publisher and subscriber that can be used to send/receive messages.

